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IKP of 6 DoF Manipulator

Given

$$H_6^0 = \begin{bmatrix} R_6^0 & (\vec{O}_0\vec{O}_6)^0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4} \quad (1)$$

$$\vec{P} = (\vec{O}_0\vec{O}_6)^0 = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (2)$$

$$R_6^0 = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad (3)$$

Wanted

$$\theta_1, \theta_2, \dots, \theta_6 \quad (4)$$

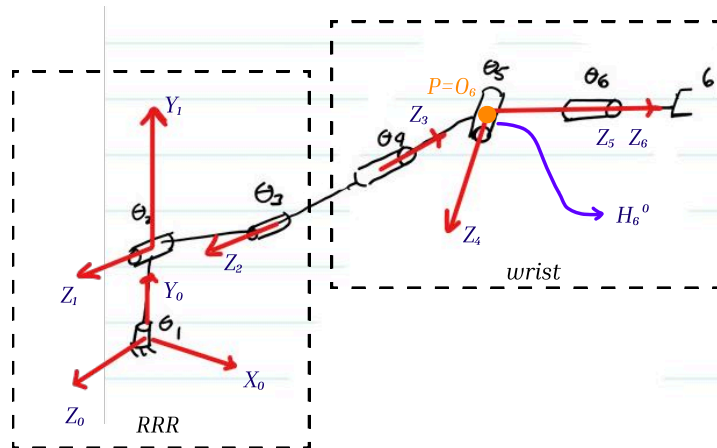


Figure 1: Schematic view of 6 DoF manipulator with coordinate frames.

This exercise focuses on a **decoupled** 6 DoF serial manipulator consisting of an *RRR* and a *Wrist* mechanism. This means the position of the end-effector is controlled using the first 3 links (θ_1 to θ_3) while the orientation of the end-effector can be adjusted by the next 3 links (θ_4 to θ_6) independently. Thus, one can solve the IKP of this manipulator by considering the first and second segments separately. Note that for simpler relations, the position of the end effector (O_6) is assumed to be attached to the wrist point P depicted in figure 1.

According to Lecture 6

$$\theta_1 = \text{atan2} \left(\frac{x}{\sqrt{x^2 + z^2}}, \frac{-z}{\sqrt{x^2 + z^2}} \right) \quad (5)$$

$$\vec{P}_1^1 = H_0^1 \vec{P}_0^0 = [X \ Y \ 0]^T \quad (6)$$

$$[H_0^1] = [R_{y_1, -\theta_1}] [T_{y_1, -h}] \quad (7)$$

$$l'_3 = l_3 + l_4 \quad (8)$$

$$l^2 = l_2^2 + l'_3{}^2 + 2l_2 l'_3 \cos \theta_3 = X^2 + Y^2 \quad (9)$$

$$\theta_3 = \cos^{-1} \frac{X^2 + Y^2 - l_2^2 - l'_3{}^2}{2l_2 l'_3} \rightarrow \begin{cases} \theta_3^1 \\ \theta_3^2 \end{cases} \quad (10)$$

$$\theta_2 = \beta - \alpha \quad (11)$$

$$\Delta O_1 P A \Rightarrow \beta = \text{atan2} \left(\frac{X}{\sqrt{X^2 + Y^2}}, \frac{Y}{\sqrt{X^2 + Y^2}} \right) \quad (12)$$

$$\Delta O_1 P B \Rightarrow \alpha = \text{atan2} \left(\frac{l_2 + l'_3 \cos \theta_3}{\sqrt{X^2 + Y^2}}, \frac{l'_3 \sin \theta_3}{\sqrt{X^2 + Y^2}} \right) \quad (13)$$

Following equations 5 to 13 gives a pair of possible answers for θ_2 and θ_3 .

Now to obtain $(\theta_4 \text{ to } \theta_6)$ we know the orientation from base to end-effector (R_0^6). Also, orientation from base to wrist link (link 3) is R_0^3 (computed from $\theta_1, \theta_2, \theta_3$). Thus, wrist orientation would be

$$R_0^3 = [R_{y_0, \theta_1}] [R_{z_1, \theta_2}] [R_{z_2, \theta_3}] \quad (14)$$

$$R_3^6 = (R_0^3)^T \cdot R_0^6 \quad (15)$$

Now having R_3^6 one can map it to the rotation matrix corresponding to successive Z-Y-Z Euler angles $(\theta_4, \theta_5, \theta_6 = \phi, \theta, \psi)$.

$$R_{(\phi, \theta, \psi)} = \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix} = R_3^6 \quad (16)$$

To find a solution for this problem we break it down into two cases. First, suppose that not both of r_{13} , r_{23} are zero. Then from Equation (2.26) we deduce that $s_\theta \neq 0$, and hence that not both of r_{31} , r_{32} are zero. If not both r_{13} and r_{23} are zero, then $r_{33} \neq \pm 1$, and we have $c_\theta = r_{33}$, $s_\theta = \pm \sqrt{1 - r_{33}^2}$ so

$$\theta = \text{Atan2} \left(r_{33}, \sqrt{1 - r_{33}^2} \right) \quad (17)$$

or

$$\theta = \text{Atan2} \left(r_{33}, -\sqrt{1 - r_{33}^2} \right) \quad (18)$$

If we choose the value for θ given by Equation 17, then $s_\theta > 0$, and

$$\begin{aligned}\phi &= \text{Atan2}(r_{13}, r_{23}) \\ \psi &= \text{Atan2}(-r_{31}, r_{32})\end{aligned}\tag{19}$$

If we choose the value for θ given by Equation 18, then $s_\theta < 0$, and

$$\begin{aligned}\phi &= \text{Atan2}(-r_{13}, -r_{23}) \\ \psi &= \text{Atan2}(r_{31}, -r_{32})\end{aligned}\tag{20}$$

Thus, there are two solutions depending on the sign chosen for θ .

If $r_{13} = r_{23} = 0$, then the fact that R is orthogonal implies that $r_{33} = \pm 1$ and that $r_{31} = r_{32} = 0$. Thus, R has the form

$$R = \begin{bmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & \pm 1 \end{bmatrix}\tag{21}$$

If $r_{33} = 1$, then $c_\theta = 1$ and $s_\theta = 0$, so that $\theta = 0$. In this case

$$R = \begin{bmatrix} \cos(\phi + \psi) & -\sin(\phi + \psi) & 0 \\ \sin(\phi + \psi) & \cos(\phi + \psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}\tag{22}$$

Thus, the sum $\phi + \psi$ can be determined as

$$\phi + \psi = \text{Atan2}(r_{11}, r_{21}) = \text{Atan2}(r_{11}, -r_{12})\tag{23}$$

If $r_{33} = -1$, then $c_\theta = -1$ and $s_\theta = 0$, so that $\theta = \pi$. In this case Equation (2.26) becomes

$$\begin{bmatrix} -c_{\phi-\psi} & -s_{\phi-\psi} & 0 \\ s_{\phi-\psi} & c_{\phi-\psi} & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & 0 \\ 0 & 0 & -1 \end{bmatrix}\tag{24}$$

The solution is thus

$$\phi - \psi = \text{Atan2}(-r_{11}, -r_{12})\tag{25}$$

As before, there are infinitely many solutions. Note that it was assumed:

$$\begin{aligned}\phi &\triangleq \theta_4 \\ \theta &\triangleq \theta_5 \\ \psi &\triangleq \theta_6\end{aligned}$$